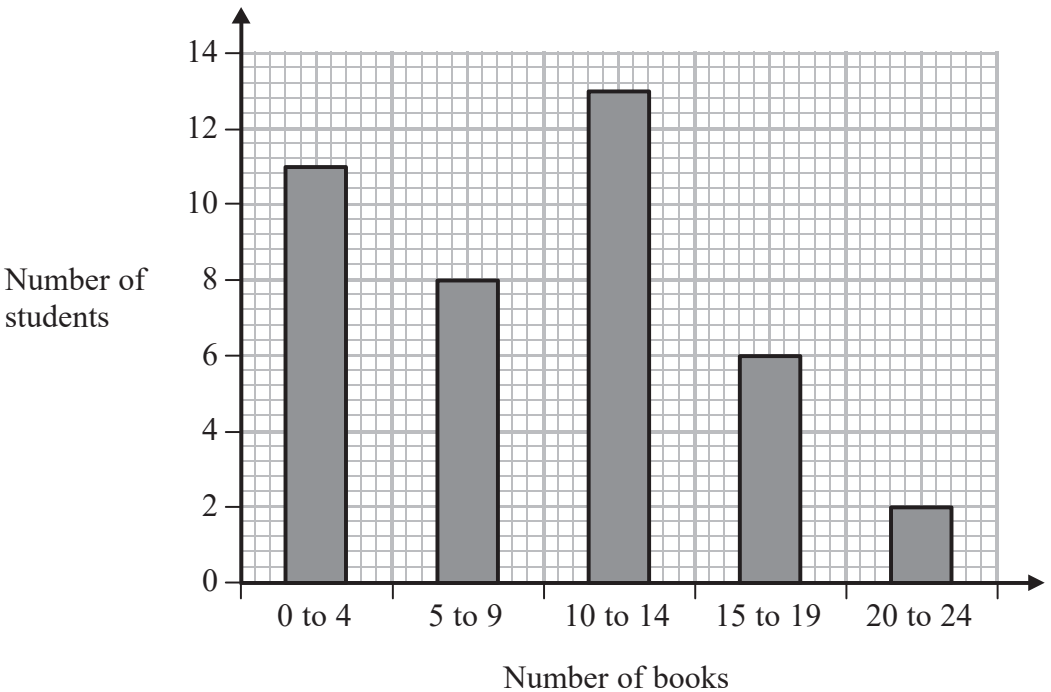
23 Fran asks each of 40 students how many books they bought last year.

The chart below shows information about the number of books bought by each of the 40 students.



(a) Work out the percentage of these students who bought 20 or more books.

2 students bought 20 or more books.

	Students	Percent	
÷20	40	100	
	2	5	÷20

1 mark
Scale 2 out of 40 to a percentage

5 ..... %

(2)

1 mark
5% of the students

(b) Show that an estimate for the mean number of books bought is 9.5
You must show all your working.

Books	Frequency f	Midpoint x	Product fx
0–4	11	2	22
5–9	8	7	56
10–14	13	12	156
15–19	6	17	102
20–24	2	22	44
Totals:	40		380

1 mark
Read frequencies and midpoints

1 mark
Calculate the products

$$\text{Estimated mean} = \frac{380}{40} = 9.5$$

1 mark
Total fx ÷ total frequency

1 mark
Correct figures show the estimate is 9.5

(4)

(Total for Question 23 is 6 marks)